

Quiz #5

K/U	A	T	C	Total
18	0	0	0	18

Name:

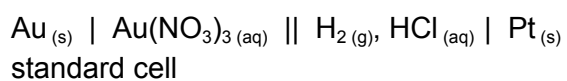
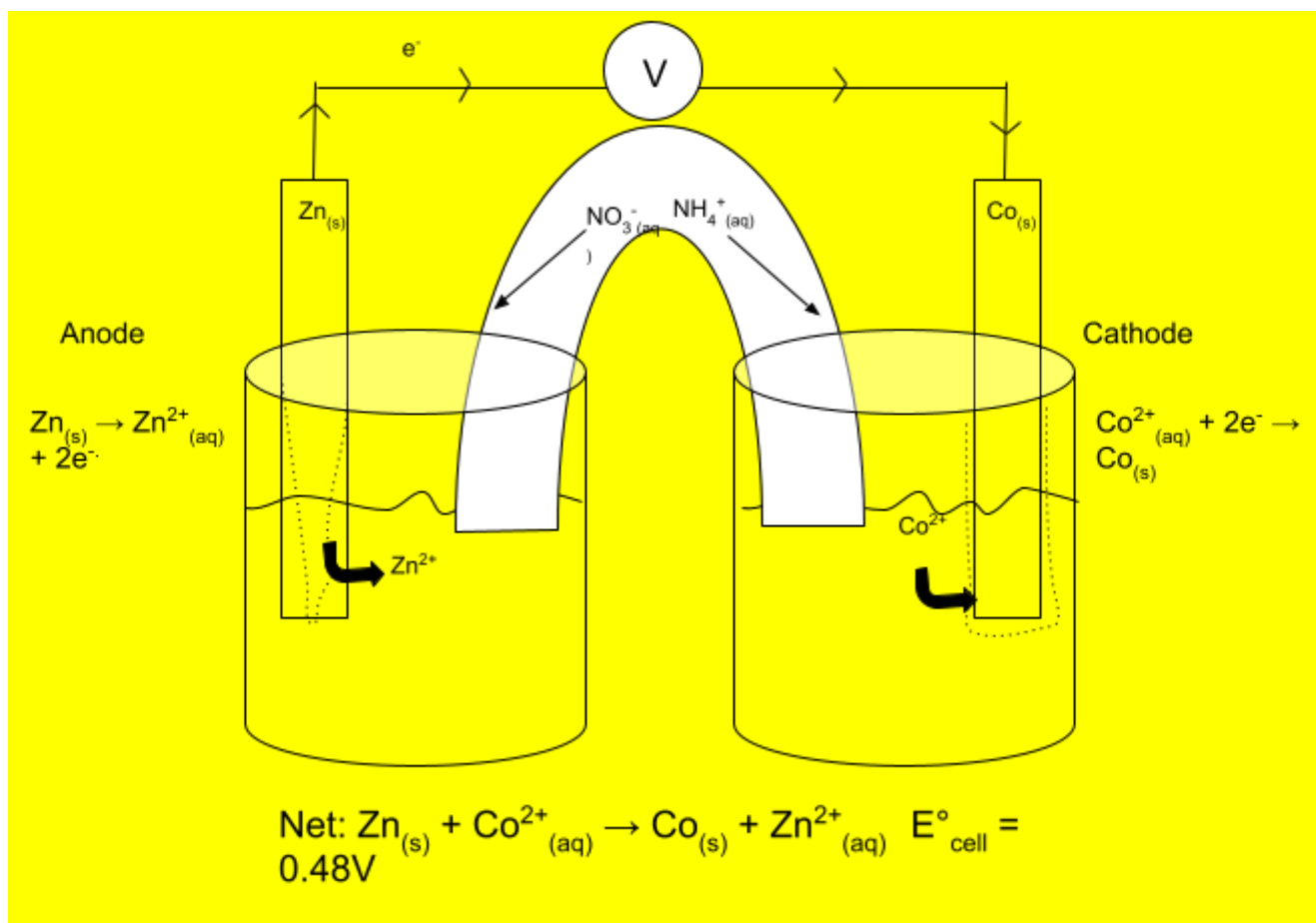
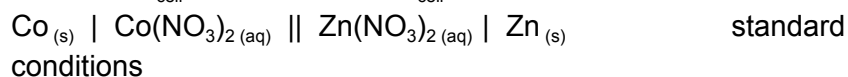
1. Draw the following galvanic cells (salt bridge design) and label the following:
 - i. anode and cathode
 - ii. oxidation and reduction half-reactions
 - iii. direction of electron flow,
 - iv. direction of cation and anion movement
 - v. net reaction
 - vi. calculate the E°_{cell} or calculate the E_{cell} if not at standard conditions
 - a. $\text{Co}_{(\text{s})} \mid \text{Co}(\text{NO}_3)_2_{(\text{aq})} \parallel \text{Zn}(\text{NO}_3)_2_{(\text{aq})} \mid \text{Zn}_{(\text{s})}$ standard conditions
 - b. $\text{Au}_{(\text{s})} \mid \text{Au}(\text{NO}_3)_3_{(\text{aq})} \parallel \text{H}_{2(\text{g})}, \text{HCl}_{(\text{aq})} \mid \text{Pt}_{(\text{s})}$ hydrogen standard cell
 - c. $\text{Sn}_{(\text{s})} \mid \text{Sn}(\text{NO}_3)_2_{(\text{aq})} (1.00 \text{ M}) \parallel \text{Zn}(\text{NO}_3)_2_{(\text{aq})} (0.0100 \text{ M}) \mid \text{Zn}_{(\text{s})}$

Draw the following galvanic cells (salt bridge design) and label the following:

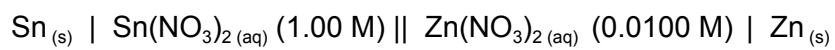
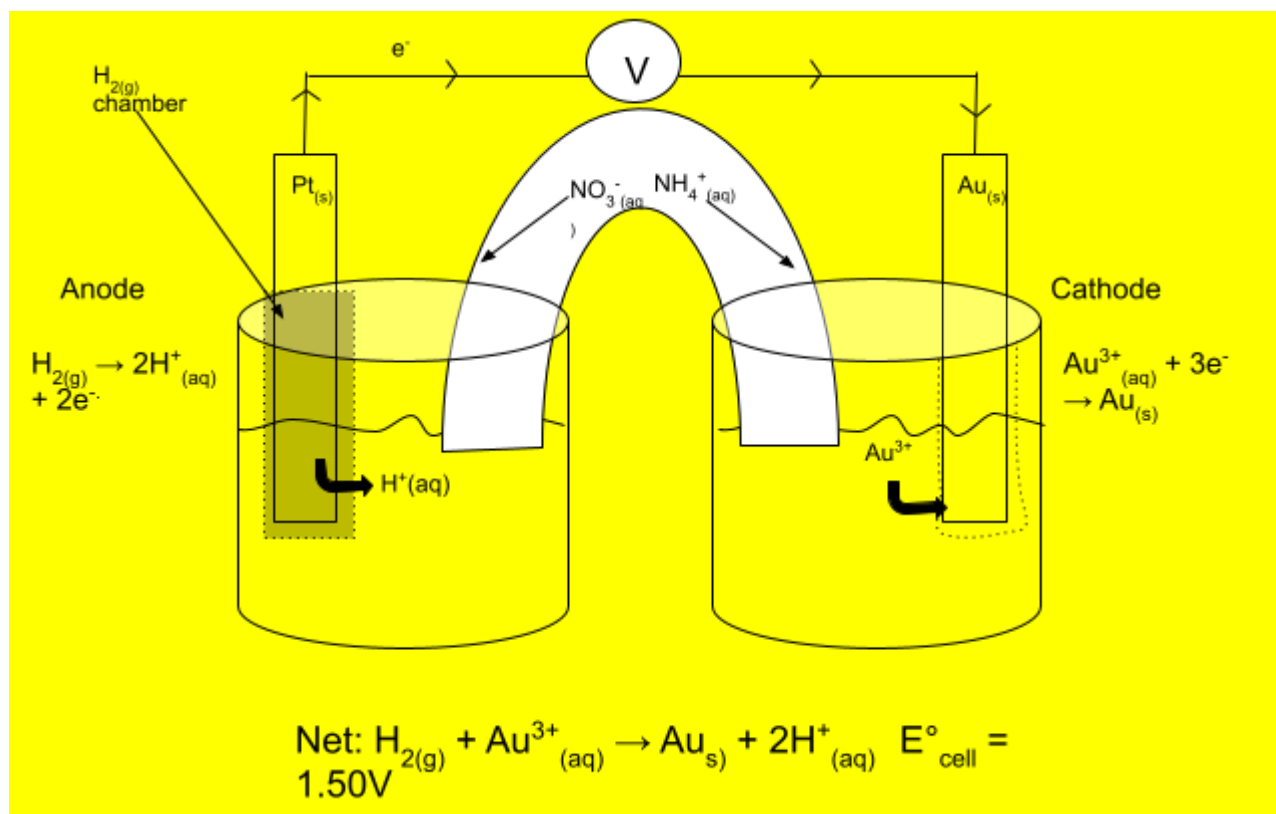
- i) anode and cathode
- ii) oxidation and reduction half-reactions
- iii) direction of electron flow
- iv) cation and anion movement

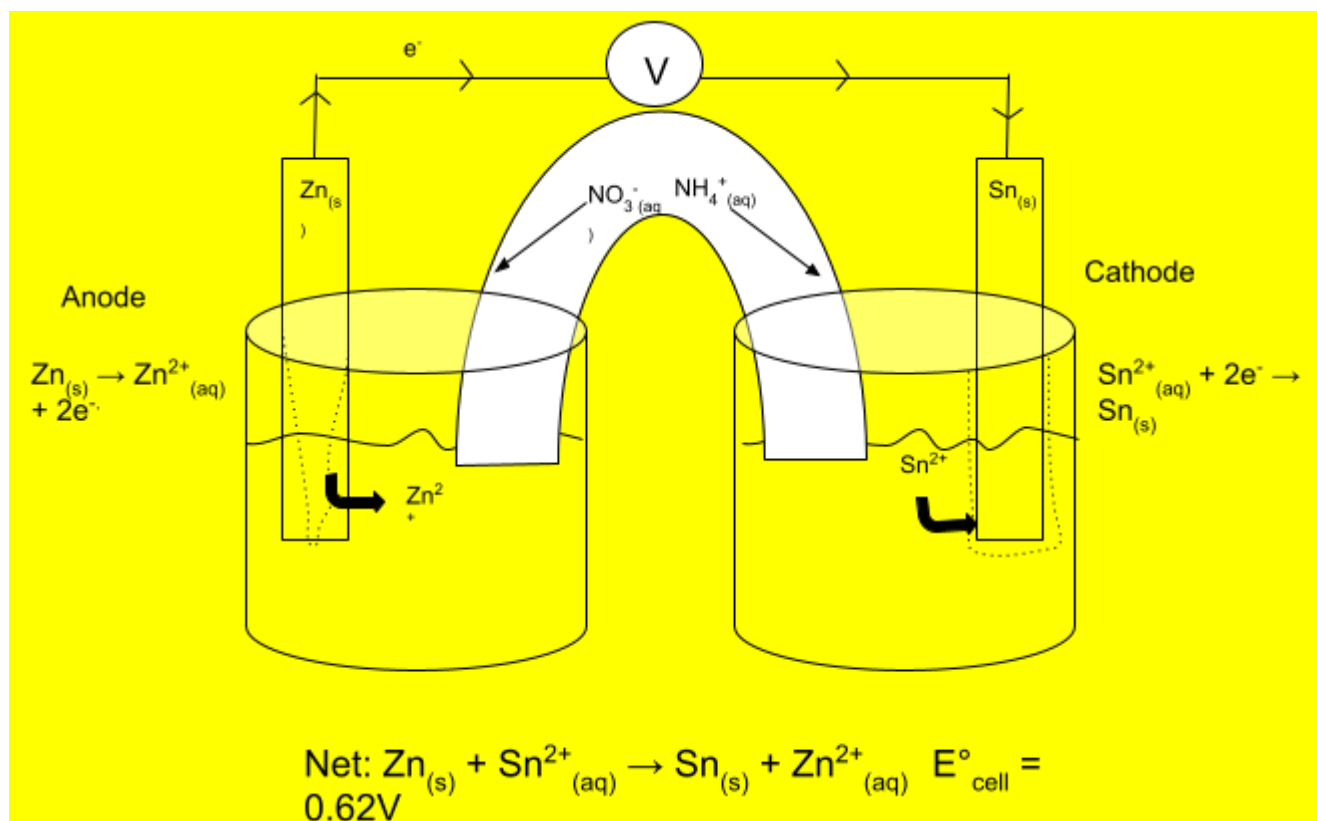
v) net reaction

vi) calculate the E°_{cell} or calculate the E_{cell} if not at standard conditions



hydrogen





$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.05916}{n} \log K^* \quad K^* = \frac{[\text{Zn}^{2+}]}{[\text{Sn}^{2+}]} = \frac{(0.0100\text{ M})}{(1.00\text{ M})} = 0.01 ; \log K = -2 \quad n = 2 \text{ mol } e^- \therefore E_{\text{cell}} = 0.68\text{ V}$$